

Marqo

Documentation Infrastructure Audit Report

Prepared by VeriOps

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Risk: High

Key Metrics

Pages scanned: 148

Report type: Premium Executive Audit

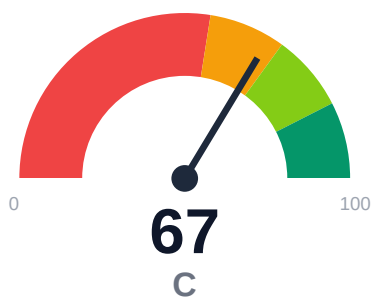
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 12
- API usage-doc coverage is low: 3.63% (64 API pages detected)
- Example reliability estimate is low: 12.41%
- Freshness visibility is weak: only 5.41% pages show last updated

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Documentation audit covered 148 pages across 2 site URLs with 100% crawl coverage and 91.5% overall confidence. Critical issues identified: 12 confirmed broken internal links affecting user navigation, API reference coverage at only 3.63% despite 64 API pages detected, and example code reliability estimated at 12.41%. Freshness metadata present on only 5.41% of pages, limiting maintainability visibility. Multi-project topology confirmed with no fragmentation penalties applied.

External posture: SEO/GEO issue rate **0.3%**, public API coverage **3.6%**, broken links **12**.

67 Audit Score	C Grade	High Risk Band
148 Pages Scanned	12 Broken Links	0.3% SEO Issue Rate

Per-Site Breakdown

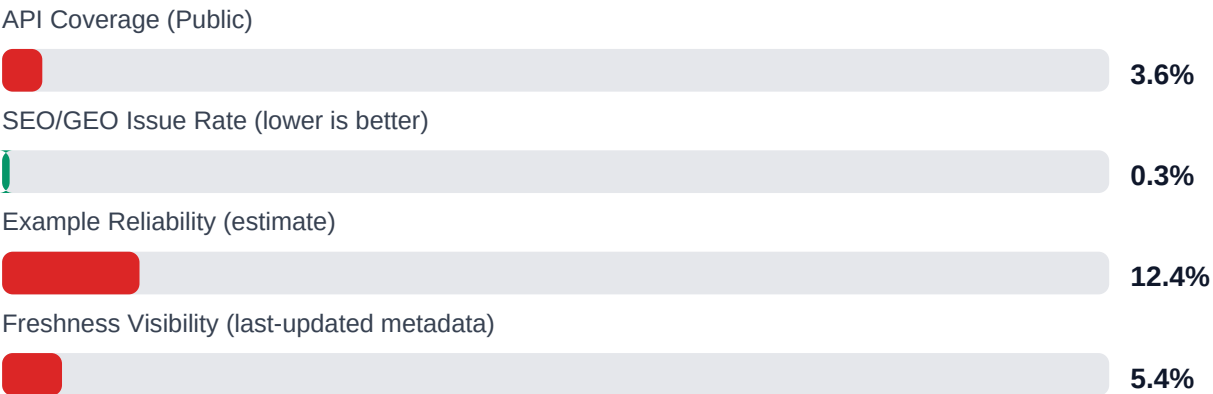
Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://docs.marqo.ai/	74	6	3.6	12.4	0.3
https://docs.marqo.ai/merchandising	74	6	3.6	12.4	0.3

Industry Benchmark Comparison

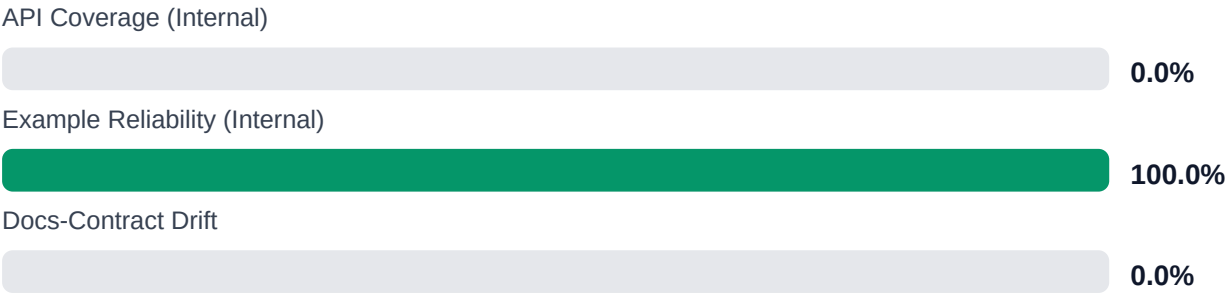
Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-25
Industry average (developer docs)	85	-18
Minimum acceptable	70	-3
Your score	67	-

Gap to industry average: **18 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics



Internal Metrics (from scorecard)



Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
\$140,822 annual exposure Remediation: \$2,090 Monthly loss: \$4,125 Opportunity: \$7,436/mo	\$249,513 annual exposure Remediation: \$5,445 Monthly loss: \$12,903 Opportunity: \$7,436/mo	\$431,728 annual exposure Remediation: \$8,800 Monthly loss: \$27,808 Opportunity: \$7,436/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	\$1,782	\$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	\$1,122	\$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	\$2,257	\$990
F-API-COVERAGE	Undocumented API operations	LOW	\$2,904	\$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	\$2,244	\$935
F-DRIFT	Code/docs contract drift	LOW	\$1,901	\$660
F-TERMINOLOGY	Terminology inconsistency	LOW	\$693	\$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. Fix all 12 confirmed broken internal documentation links to restore navigation integrity [impact: critical, effort: low]

2. Expand API reference documentation from 3.63% to minimum 60% coverage across 64 detected API pages [impact: critical, effort: high]
3. Audit and update example code to improve reliability from 12.41% to target 80%+ [impact: critical, effort: high]
4. Implement last-updated metadata across all pages to increase freshness visibility from 5.41% to 100% [impact: high, effort: medium]
5. Resolve SEO/geo issues affecting 0.34% of pages to optimize search discoverability [impact: medium, effort: low]

Expert Analysis: Strengths and Risks

Strengths

- + Complete crawl coverage achieved at 100% (148 of 148 discovered pages)
- + High overall audit confidence at 91.5% with strong link verification (95% confidence)
- + Zero repository navigation broken links; all 12 broken links are user-facing documentation issues
- + Multi-project structure properly configured without fragmentation penalties
- + No unverified or excluded link issues detected

Risks

- 12 confirmed broken internal documentation links directly impacting user navigation and trust
- API reference coverage critically low at 3.63% with 64 API pages detected, indicating severe documentation gaps
- Example code reliability at 12.41% suggests majority of code samples may be outdated or non-functional
- Freshness metadata missing on 94.59% of pages, preventing users from assessing content currency
- SEO/geo issues present at 0.34% rate, potential discoverability impact
- Low example reliability combined with low API coverage creates compounding onboarding friction
- Absence of last-updated timestamps undermines documentation credibility for enterprise evaluation

Limitations

- * External audit cannot verify functional correctness of API documentation content or parameter accuracy
- * Example code reliability is estimated through heuristics; actual runtime testing required for true validation
- * Crawl limited to publicly accessible pages; authenticated or gated content not evaluated
- * API coverage detection may undercount if non-standard documentation patterns used; 85% confidence indicates possible detection gaps
- * Freshness assessment relies on explicit metadata presence; actual content currency not semantically analyzed

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.